

Exact Observations in Copy-Discard Categories

Abstract for the talk at the next retreat

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This talk presents an axiomatisation of exact observations in copy-discard categories and proposes a class of semantic universes for probabilistic programs with exact observations. The program below infers the mean of a normally distributed random variable; the program is written in `do` notation but features an additional `observe` construct, which we seek to give semantics to.

```
do( $m$ )  
   $x_1 \leftarrow \mathcal{N}(m, 1)$   
  observe( $x_1 = 3$ )  
   $x_2 \leftarrow \mathcal{N}(m, 1)$   
  observe( $x_2 = 2.5$ )  
return( $m$ )
```

We propose partial Markov categories with exact observations as a categorical semantics for such programs. We prove a synthetic Bayes theorem for partial Markov categories with exact observations and investigate other properties of this structure. Given a Markov category with conditionals, $\mathbb{C}$, we construct the free partial Markov category with exact observations on it, $\text{exOb}(\mathbb{C})$, and show that morphisms in $\text{exOb}(\mathbb{C})$ have a normal form which is probably not unique.

Future work. We would like to investigate the normal form of morphisms in $\text{exOb}(\mathbb{C})$: is there a quotient that makes it unique? We would also like to investigate the relationship between the `exOb` construction and Markov magmoids.

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